Live Tracker: what the backend needs to build

Project: Workforce Pro, Attendance & HR (Box For Less, Double A)
Date: 12 November 2026
For: the backend developer
Replaces: §3, §5 (tracking tables) and §8 of backend-live-tracker-and-sms-verification.md. The SMS parts of that document still apply, with the distance rules from backend-multi-shop-and-places.md.

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1. The short version

The Live Tracker lets HR see **where field staff are while they are on duty**: a live map and a daily report. The frontend is finished. The server side does not exist yet, so today the tracker works only inside one browser. Locations a phone records are saved in that phone's `localStorage`, and an HR user on another computer never sees them.

What the backend needs to do:

1. When a field employee **clocks in**, open a tracking session and give the phone a **tracking token**.
2. Accept the phone's **location points** while the session is open, and refuse them after clock-out.
3. Give HR the **latest position of each field employee** for the live map (every 15 seconds, or pushed through Supabase Realtime).
4. At the end of each day, **summarise the trail** (time covered, distance, shop stays), save the summary, and **delete the points**.
5. Serve the **daily report** from those summaries.

2. Who is tracked, and when

Tracked	Never tracked
Supervisors, Sales Representatives, Merchandisers	Warehouse Workers, office staff, HR

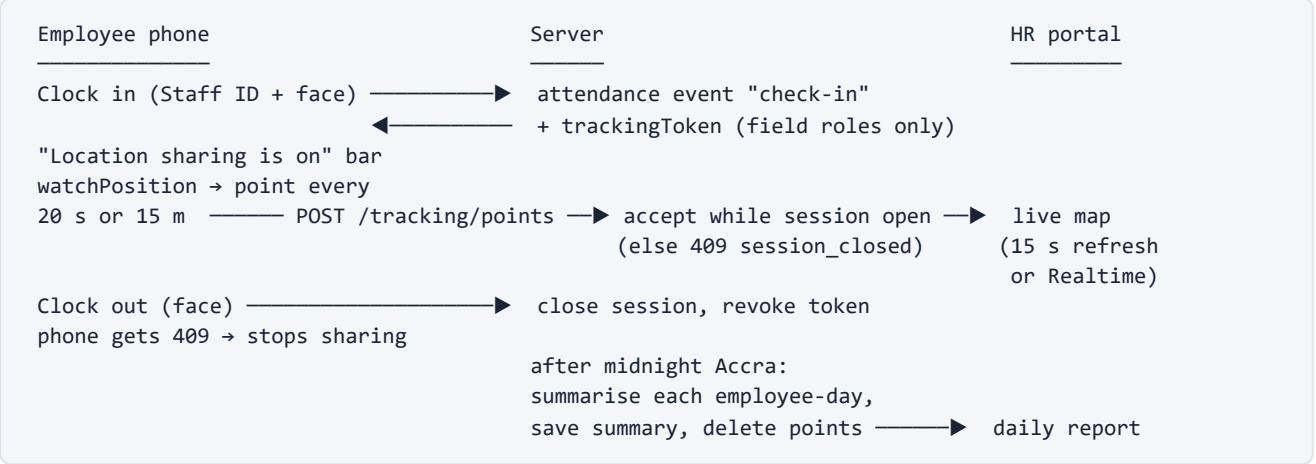
The role comes from the job position: `categoryForPosition()` in `src/constants/employeeCategories.js` returns `supervisor`, `sales-rep` or `merchandiser`.

Rules the server must enforce

1. **Only between clock-in and clock-out.** A point recorded before clock-in or after clock-out is rejected, even if it arrives later.
2. **Server time decides.** Clock-in, clock-out and `receivedAt` use server time. The phone's `recordedAt` is stored, but only checked against the session window.
3. **Company separation.** An HR user sees only their own company's people, through RLS on every table below.
4. **A lost signal is never misconduct.** "Signal lost" and low coverage are things for HR to ask about. They never change attendance.

5. **No positions are kept after the day.** The Tracking Policy shown to staff (`/tracking-policy`) promises that the detailed trail is deleted at the end of the working day, and only a summary without coordinates is kept.
6. **Distance only, HR decides.** There is no geofence or pass/fail on distance. HR sees "40 m" or "2.3 km" and judges. The one fixed number is **100 m**, an allowance for GPS error when measuring a shop stay (§6).

3. How it works, start to finish



3.1 On the phone (already built)

- After a field employee clocks in, the phone shows a bar saying **"Location sharing is on"** (`src/components/tracking/LocationSharingBar.jsx`). It uses `navigator.geolocation.watchPosition` with high accuracy.
- It keeps a point only when **20 seconds** have passed **or** the phone has moved **15 m** since the last kept point, up to **2,000 points per person per day** (`src/services/locationTrackingService.js`).
- The bar tells the employee when something is wrong: no mobile data, location permission blocked, GPS off.
- The phone that did the clock-in is the phone that shares location. It stops by itself at clock-out.
- **Limitation:** a web app can only send locations while it is open on screen. When the phone is locked or the employee switches apps, points stop and HR sees "signal lost". A native wrapper (Capacitor with a background-location plugin) is planned later and will use the same endpoints, sending queued batches when it is back online.

3.2 Tracking session

- On a successful **check-in** for a field role, create a `tracking_sessions` row and return a `trackingToken` in the `POST /attendance/events` response. Return nothing for other roles.
- The token is random, stored **hashed**, bound to that employee and that day's session, and usable only for `POST /tracking/points` .
- On **check-out**, close the session and revoke the token.
- Also close it when the employee is marked absent (face scan locked out), and at the end of the working day if there was no clock-out (attendance already flags "no clock-out").

4. Endpoints

4.1 `POST /tracking/points` (phone)

`Authorization: Bearer <trackingToken>`

```
{
  "points": [
    { "latitude": 5.60371, "longitude": -0.18696, "accuracy": 12,
      "recordedAt": "2026-11-12T09:41:07Z", "isMock": false }
  ]
}
```

- Accept a **batch**: the web app sends one point at a time, while the native app will send queued batches.
- Reject a point if `recordedAt` is before clock-in, after clock-out, more than 2 minutes in the future, or more than 12 hours old.
- Store `receivedAt` (server time) with each point.
- Apply the same thinning as the phone (20 s or 15 m) and the 2,000-per-day cap, so a modified phone cannot flood the table.
- **Flag, don't drop:**
  - points with `isMock: true` (Android reports fake-location apps);
  - jumps that imply more than 200 km/h.

Raise a high-severity alert for HR in either case.

- Response: `{ "accepted": 3, "rejected": 1, "sessionOpen": true }`.
- After clock-out: `409 { "code": "session_closed" }`. The phone stops sharing when it gets this.
- Rate-limit by token.

4.2 Live map data (HR)

The HR page needs, per company, **one entry per field employee** in the shape `getLiveTracker()` returns today (`src/services/liveTrackerService.js`). The page refreshes every **15 seconds** and pauses when the browser tab is hidden.

```
{
  "asOf": "2026-11-12T10:02:00Z",
  "people": [
    {
      "employeeId": "BFL-0007",
      "fullName": "…",
      "category": "sales-rep",
      "duty": "Clocked in — live",
      "onDuty": true,
      "visitsToday": 3,
      "lastEvent": "Live location",
      "lastEventAt": "2026-11-12T10:01:40Z",
      "position": { "latitude": 5.60, "longitude": -0.18, "accuracy": 12,
                    "source": "live", "at": "2026-11-12T10:01:40Z", "place": null },
      "shops": [{ "shopId": "shop-bfl-001", "shopName": "…", "latitude": 5.61, "longitude": -0.19 }]
    }
  ],
  "counts": {
    "supervisor":   { "total": 2, "placed": 1, "onDuty": 1 },
    "sales-rep":    { "total": 5, "placed": 4, "onDuty": 4 },
    "merchandiser": { "total": 9, "placed": 7, "onDuty": 8 }
  }
}
```

- `shops` lists the person's 1 to 7 assigned shops (for Sales Reps and Supervisors, their daily list). The map uses it to show **"Nearest shop — 40 m away"**.
- The page also draws all the company's shops on the map, which it already reads from the `shops` table.

You can serve this in either of two ways:

- **(A) An endpoint**, `GET /hr/tracking/live?companyId=`, that builds the JSON above.
- **(B) Tables plus RLS**, like the rest of the HR portal (`src/services/backend/cache.js`):
  - a view `tracking_latest (staff_id, company_id, latitude, longitude, accuracy, recorded_at, session_status)` with one row per employee, today only;
  - added to Realtime.

The frontend then builds the JSON itself. **Option B is preferred:** it matches how HR already reads data, and Realtime makes the map truly live.

Tell me which one you choose and I will switch `getLiveTracker()` to it.

Position and status rules (keep these identical to the frontend):

Employee state	`position`	`duty`
Clocked in, latest point at most **5 min** old	that point, `source: "live"`	`Clocked in — live`
Clocked in, latest point older than 5 min	that point, `source: "live"`	`Clocked in — signal lost N min ago`
Clocked in, no point yet	Sales Rep / Supervisor: the latest visit's GPS (`gps`) or the visited shop (`outlet`); Merchandiser: their shop (`shop`)	`Clocked in — no live location yet`
Clocked out	`null` (taken off the map)	`Clocked out — tracking stopped`
Not clocked in, absent, on leave	`null`	`Not clocked in` , `Marked absent` , `On leave`

On the map, people last seen more than 60 minutes ago are faded. This happens in the frontend; the server doesn't need to do it.

4.3 Daily report (HR)

`GET /hr/tracking/reports/daily?companyId=&date=2026-11-12&role=` returns one row per field employee, in the shape of `trackingDailyReport()` in `src/services/trackingReportService.js`:

```
{
  "dateKey": "2026-11-12",
  "rows": [{
    "employeeId": "BFL-0007", "fullName": "…", "category": "sales-rep",
    "clockIn": "…", "clockOut": null, "stillClockedIn": true,
    "visits": 4,
    "trackedMinutes": 182, "coverage": 0.93, "signalLost": 1, "longestGapMinutes": 9,
    "distanceKm": 12.4, "points": 311,
    "shops": { "listed": 4, "proven": 3, "flagged": 1, "missed": 0 },
    "result": "good"
  }],
  "totals": { "staff": 9, "clockedIn": 7, "tracked": 6, "averageCoverage": 0.88,
              "distanceKm": 61.3, "noSignal": 1 }
}
```

- `result`:
  - `good`: coverage of at least 90%.
  - `partial`: at least 60%.
  - `poor`: below 60%.
  - `no-signal`: clocked in, but no points.
  - `not-clocked-in`.
- Today's report is calculated live up to the current time. Past days come from the saved summaries.
- The report contains **no coordinates**.
- The CSV export already happens in the browser and is recorded through `/hr/audit/exports`. No new endpoint is needed for it.

5. End of day: summarise, then delete

Run a job after midnight Africa/Accra, and on demand for any day not yet processed. For each employee-day:

Field	Meaning
`clockIn`, `clockOut`	from attendance
`trackedMinutes`	clock-out − clock-in
`points`, `firstAt`, `lastAt`	how many points, and the first and last times
`coveredMinutes`, `coverage`	each point covers the time until the next point, **up to 5 minutes**; coverage = covered ÷ tracked (0 to 1)
`signalLost`, `longestGapMinutes`	gaps longer than 5 minutes between clock-in, the points and clock-out

Field	Meaning
`distanceM`	sum of distances between consecutive points, **leaving out points with accuracy worse than 100 m**
`shopStays`	`{ shopId: { minutes, longestMinutes, closestM } }`, see §6

Save the summary, then **delete that day's points**. The reference implementation is `summariseTrail()` in `locationTrackingService.js`; the tests in §8 must give the same numbers.

6. Shop stays (proof of a shop visit)

Some Sales Reps and Supervisors have claimed shop visits without going. The tracker is what proves a visit:

- A **stay** at a shop is a run of consecutive points **within 100 m** of the shop, with no gap longer than **5 minutes**. Its length runs from the first point to the last.
- A listed shop is **proven** when its longest stay is **at least 10 minutes** (`MIN_STAY_MINUTES`).
- The reference implementation is `shopStays()` and `visitProof()` in `src/services/visitListService.js`.

Result	Rule
`proven`	Stay of 10+ minutes within 100 m
`no-stay`	A visit was recorded, but no 10-minute stay shows on tracking (flag for HR)
`unmeasured`	A visit was recorded, but the shop has no map location (HR should add one)
`pending`	Nothing yet, and the day is not over
`missed`	Nothing, and the working day is over (flag for HR)

Visits to shops that aren't on the person's list are flagged `off-list`, and typed-in outlets are flagged `not-in-list`. Neither ever counts as proven.

Store `shopStays` in the summary **before** deleting the points.

7. Tables

Table	Columns
`tracking_sessions`	`id, staff_id, company_id, date, clock_in_at, clock_out_at, token_hash, status ('open'|'closed'), closed_reason`
`location_points`	`id, session_id, staff_id, company_id, latitude, longitude, accuracy, recorded_at, received_at, is_mock, flagged` (**deleted after the day's summary**)
`tracking_day_summaries`	`staff_id, company_id, date, clock_in, clock_out, tracked_minutes, points, first_at, last_at, covered_minutes, coverage, signal_lost, longest_gap_minutes, distance_m, shop_stays jsonb` (**no coordinates**)

Access:

- **RLS:** HR users with the `sales-monitoring` permission can read their own companies' rows.
- **Writes:** only the edge function writes. Phones never write directly.
- **Indexes:** `location_points (staff_id, recorded_at)` and `tracking_sessions (staff_id, date)`.

8. Acceptance checks

Session window

- A point recorded 1 second after clock-out is rejected. One recorded 1 second before is accepted, even if it is uploaded later.
- After clock-out, `POST /tracking/points` returns `409 session_closed`.
- A Warehouse Worker or office employee never gets a tracking token.

Live map

- A Box For Less HR user cannot read Double A points, sessions or summaries, and vice versa.
- The latest point is 6 minutes old: duty is "signal lost 6 min ago".
- After clock-out the person disappears from the map.

Coverage and distance

- 60-minute day, one point per minute for the first 30 minutes, then nothing: `coveredMinutes = 35`, `coverage ≈ 0.58`, `signalLost = 1`, `longestGapMinutes = 30`.
- Clocked in with no points: `coverage = 0`, `result = no-signal`.
- A point with 500 m accuracy is left out of the distance.

Shop stays

- Points every 3 minutes within 100 m of a shop from 09:00 to 09:12: stay = 12 minutes, so the shop is `proven`.
- Points at 0, 4, 15 and 19 minutes: the 11-minute gap breaks the stay, so the longest stay is 4 minutes and the shop is not proven.

After the end-of-day job

- That day's points are gone.
- The summary has no latitude or longitude.
- The stay minutes are still there.

9. Frontend functions this replaces

Frontend today (browser-only)	Replace with
`locationTrackingService.recordLocationPing()`	`POST /tracking/points`
`trackingStatus()`, `onAttendanceRecorded()`	`trackingToken` in the check-in response; `409 session_closed`
`latestLocation()`, `todayTrail()`	`tracking_latest` (option B) or `GET /hr/tracking/live`
`liveTrackerService.getLiveTracker()`	built from the above
`daySummary()`, `summariseTrail()`, `dayShopStays()`	end-of-day job + `tracking_day_summaries`

Frontend today (browser-only)	Replace with
`trackingReportService.trackingDailyReport()`	`GET /hr/tracking/reports/daily`
`visitListService.getVisitList()` / `mockSaveVisitList()`	`GET/PUT /employees/:staffId/visit-list` (see the multi-shop document)

These are the browser keys to clear when testing the current version: `mock-tracking:pings`, `mock-tracking:day-summaries` and `mock-tracking:device-staff`.

10. Privacy and security

- Staff are told about tracking in the in-app Tracking Policy and Privacy Policy. The server must do what those pages promise: tracking only on duty, and the trail deleted the same day.
- Tokens are random, hashed, single-purpose and expire at clock-out.
- HTTPS only. Rate-limit the points endpoint.
- Ghana Data Protection Act, 2012 (Act 843): the company should register with the Data Protection Commission before go-live.

11. Decisions for the company

1. Whether HR needs a **past-day route** (positions). Today only summaries are kept; keeping routes would mean changing the Tracking Policy.
2. Whether the 10-minute minimum stay is right.
3. How long summaries are kept. The proposal is 6 years; see `src/constants/legal.js`.
4. When to build the **native app** for background tracking. Until then, tracking pauses whenever the app is not on screen.